

M.Tech Synopsis Report

Category	Details
Name of Institute	Shri Balasaheb Mane Shikshan Prasarak Mandal's Ashokrao Mane Group of Institutions
Name of Course	M. Tech. Computer Science Engineering
Name of Student (PRN No.)	Utkarsh Karale (PRN: [To be filled])
Date of Registration	[To be filled]
Name of Guide	[To be filled]
Type of Master's Research Project	Non-sponsored
Name of Organization / Industry	NA
Proposed Topic	PhishScope: An Explainable Hybrid Intelligence Framework for Static and Dynamic Phishing Website Analysis

1. Introduction / Background

Phishing websites have evolved into sophisticated cyberattacks that employ dynamic web content, JavaScript execution, multi-stage redirections, cloaking techniques, and rapidly changing infrastructure to evade conventional detection mechanisms. While recent machine learning and deep learning approaches have significantly improved detection accuracy, most existing solutions rely primarily on static website features such as URLs, HTML structure, domain information, and SSL certificate metadata. This research introduces PhishScope, an explainable hybrid intelligence framework that combines multi-source static feature analysis with

isolated browser-based dynamic execution to generate transparent, evidence-based risk assessments.

2. Relevance / Motivation

Static-only analysis often fails to capture malicious runtime behaviors revealed during website execution. Furthermore, many current machine learning detection systems operate as black-box models, providing limited interpretability. This lack of transparency restricts practical adoption in real-world cybersecurity environments. Developing an integrated framework that applies Explainable Artificial Intelligence (XAI) to both static and dynamic telemetry will improve detection reliability, foster user trust, and empower cybersecurity analysts with deployable investigation tools.

3. Literature Review

Paper Title & Year	Feature(s) Used	Model/Technique
Phishing URL Detection with Contextual Embeddings (2025)	URL	Hybrid ML
Impact of Feature Engineering in Phishing Detection (2025)	URL, HTML	Random Forest
Explainable Machine Learning for Phishing Detection (2025)	URL, Domain	XGBoost + SHAP
Cross-Dataset Analysis for Phishing (2025)	URL, HTML	LightGBM
Deep Learning-Based Website Detection (2024)	HTML	CNN

4. Literature Review Summary

The review of current literature highlights a strong dependency on static URL and HTML features for phishing detection. Most contemporary studies focus on maximizing classification accuracy using models like Random Forest, CNNs, or BERT, but heavily neglect dynamic runtime behaviors (such as JavaScript execution or redirect chains). Additionally, while a few recent papers integrate XAI (like SHAP), they typically do so over limited feature sets rather than a comprehensive, multi-source framework.

5. Research Gap

- **Limited Integration of Multi-Source Features:** Current models rarely integrate URL, HTML, Domain, and SSL data into a single pipeline.
- **Insufficient Dynamic Behavioral Analysis:** Modern phishing exploits runtime behaviors that static scanners miss entirely.
- **Lack of Explainable Decision-Making:** Prevalent "black-box" systems offer limited transparency, reducing trust.
- **Absence of Integrated Frameworks:** Focus is often placed solely on classification metrics rather than unified risk assessment systems.
- **Limited Practical Deployment:** Validations are frequently offline, lacking user-oriented tools like browser extensions.

6. Problem Statement

Phishing websites have evolved into sophisticated cyberattacks that employ dynamic web content, multi-stage redirections, and rapidly changing infrastructure. Most existing solutions rely primarily on static features and operate as black-box models, providing limited interpretability. Therefore, there is a critical need for an explainable hybrid intelligence framework that combines multi-source static analysis with isolated browser-based dynamic execution and Explainable Artificial Intelligence (XAI) to generate transparent, evidence-based risk assessments for practical cybersecurity deployment.

7. Objectives

1. To extract and integrate multi-source static website features, including URL, HTML, domain, and SSL certificate information.
2. To design and develop a hybrid intelligence framework combining static feature analysis

with dynamic browser-based behavioral analysis.

3. To monitor runtime website behaviors (JavaScript, network requests, DOM changes) within an isolated browser environment.
4. To incorporate Explainable Artificial Intelligence (XAI) techniques (SHAP/LIME) for generating transparent predictions.
5. To evaluate the framework's effectiveness, reliability, and practical applicability using standard cybersecurity metrics.

8. Proposed Methodology

The methodology follows a hybrid intelligence architecture integrating data collection, dynamic execution, model training, and XAI interpretation.

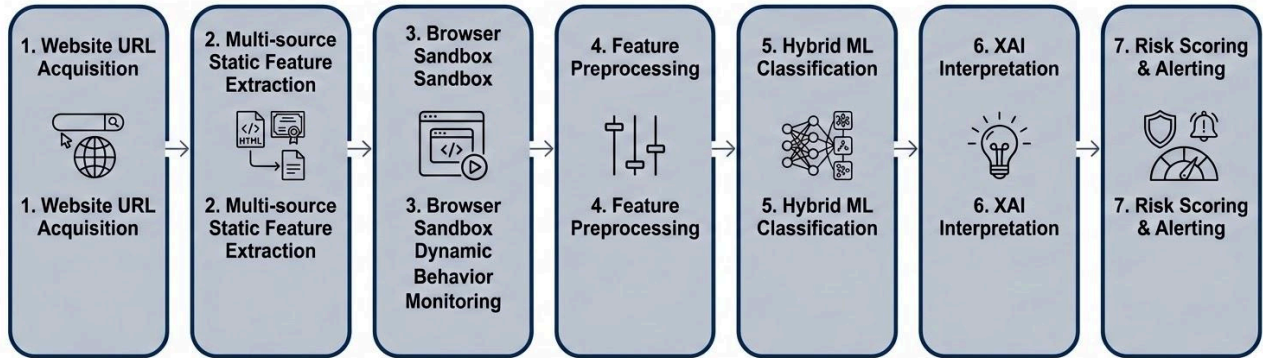
- **Design/Modelling:** A multi-layer architecture collecting URL, HTML, Domain, SSL, and DNS indicators alongside a sandbox execution module.
- **Simulation/Analysis:** Utilizing isolated browser environments to safely monitor redirect chains, DOM modifications, and browser interactions.
- **Experimental Work:** Training hybrid ML models (e.g., Random Forest + XGBoost) on processed feature sets split into 70% Train, 15% Validation, and 15% Test datasets.
- **Data Collection:** Aggregating malicious and benign instances from repositories like PhishTank and OpenPhish.
- **Validation:** Performance tracked via Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Methodology Flow

Website URL Acquisition → Multi-source Static Feature Extraction → Browser Sandbox
Dynamic Behavior Monitoring → Feature Preprocessing → Hybrid ML Classification → XAI
Interpretation → Risk Scoring & Alerting.

PhishScope Methodology Flow

End-to-End Phishing Website Detection Process



Expected Cost of the Dissertation Work

Sr. No.	Part Description	Approx. Cost (Rs)
1	Cloud Instances / Sandbox Infrastructure	10,000/-
2	Data Acquisition / API Keys	15,000/-
3	Software Licensing & Security Tools	2,000/-
4	Publication / Registration Fees	5,000/-
Total (Rs.)		32,000/-

9. Expected Outcomes

The primary outcome is the deployment of the PhishScope framework, capable of providing evidence-based, highly accurate phishing risk assessments. A secondary outcome includes developing practical cybersecurity tools, specifically a browser extension for real-time alerting and a desktop investigation application.

10. Research Contribution

This research bridges the gap between high classification performance and human understanding in cybersecurity. Academically, it presents a novel hybrid architecture integrating static telemetry with behavioral analysis and SHAP/LIME interpretation. Practically, it produces a deployable prototype framework suitable for direct industry integration and future academic expansion.

11. Scope of Work

The scope is strictly bounded to the analysis and detection of phishing websites through static indicators and isolated dynamic execution. It explicitly excludes phishing emails, SMS phishing (smishing), voice phishing (vishing), enterprise-scale threat intelligence, and deep malware reverse-engineering.

12. Work Plan

Activity	Proposed Duration
Literature Review & Research Planning	Month 1 - 2
Problem Definition & Methodology Design	Month 2 - 3
System Development & Data Collection	Month 3 - 4
Model Training & Experiments	Month 5 - 6
Performance Evaluation & Validation	Month 6 - 7
Thesis & Research Paper Writing	Month 7 - 9

13. References

- Kumar, A., et al. (2025). Phishing URL Detection with Contextual Embeddings and Machine Learning. *IEEE Conference*.

- Sharma, B., et al. (2025). Evaluating the Impact of Feature Engineering in Phishing URL Detection. *IEEE Access*.
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